

DATA PIPELINE_04: BLOB ,ADLS to SQLDB (ADF USING PowerQuery)

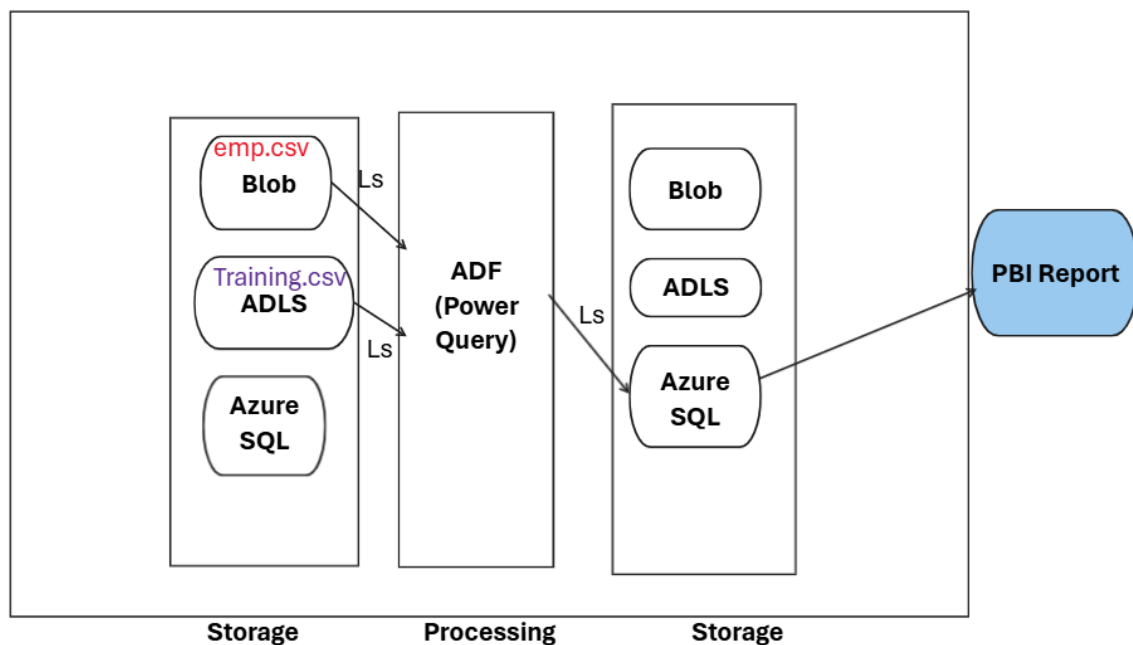
Project Objective:

To extract data from Azure Blob Storage and ADLS, transform it using Azure Data Factory Mapping PowerQuery, and load it into Azure SQL Database to create a reliable, scalable, and efficient data integration solution for analytics and reporting.

Process:

Raw Data Sources (Azure Blob Storage + ADLS) → Azure Data Factory Pipeline → Mapping PowerQuery(Transformations) → Azure SQL Database → Reporting/Analytics Layer

Architecture:

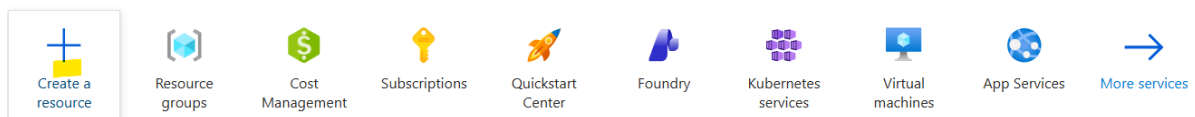


1)TO CREATE BLOB STORAGE:

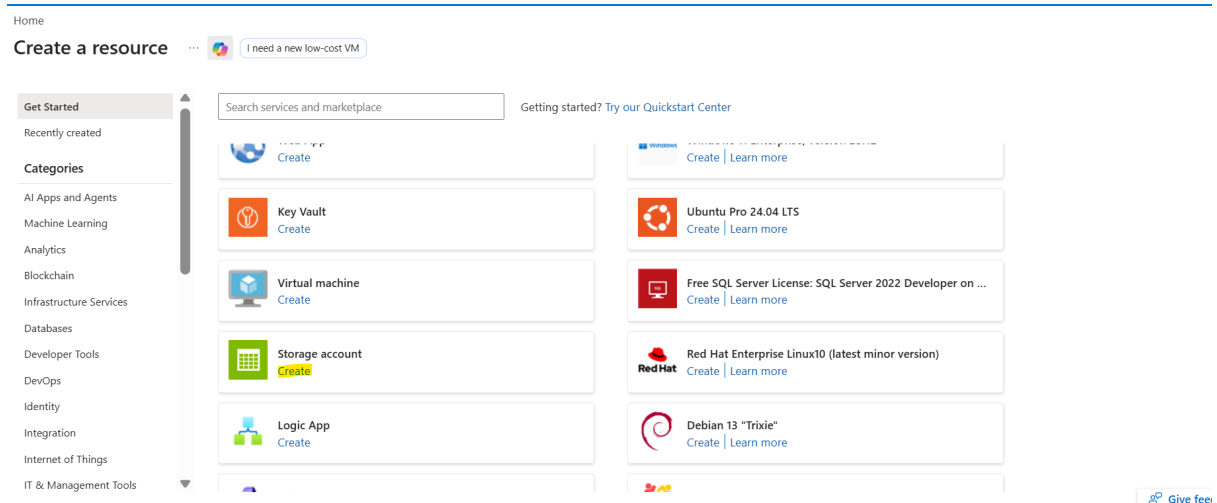
1.1) login to azure portal and signin with email id

1.2)Create Resource:

Azure services



1.3) Click on the + icon and select the storage account:



1.4) Fill all the basic details:

Subscription: Give your subscription :ex) Azure subscription1

Resource group: click create-->--new--Create a new resource group

Instance details:

Storage account name: give any name for storage account like pipelin3sa

Region: select your nearest region

Preferred storage type: Azure blob storage and azure data lake storage gen2

Performance: Standard

Redundancy: LRS depends on the data

Create a storage account ...

Subscription *	NK
Resource group *	(New) pipeline3rg Create new
Instance details	
Storage account name * ⓘ	pipeline3sa
Region * ⓘ	(US) West US 3 Deploy to an Azure Extended Zone
Preferred storage type	Azure Blob Storage or Azure Data Lake Storage i This helps us provide relevant guidance. It doesn't restrict your storage to this resource type. Learn more
Performance * ⓘ	☒ Standard: Recommended for most scenarios (general-purpose v2 account) ☐ Premium: Recommended for scenarios that require low latency.
Redundancy * ⓘ	Locally redundant storage (LRS)

Previous

Next

Review + create

Fill in the advanced details:

Blob storage:select how often we use the data depends on(HOT,COOL,COLD)

Public network access:Enable

Public network access scope:Enable all networks

Routing preference:Microsoft Network Routing

Enable soft delete:Give the no of days you can retrieve the data once it deleted

Reference:

Create a storage account

- Basics
- Advanced
- Networking
- Data protection
- Security
- Encryption
- Tags
- Review + create

[View automation template](#)

Basics

Subscription	NK
Resource group	pipeline3rg
Location	West US 3
Storage account name	pipeline3sa
Primary service	Azure Blob Storage or Azure Data Lake Storage
Performance	Standard
Replication	Locally redundant storage (LRS)

Advanced

Enable hierarchical namespace	Disabled
Enable SFTP	Disabled
Enable network file system v3	Disabled

Performance	Standard
Replication	Locally redundant storage (LRS)

Advanced

Enable hierarchical namespace	Disabled
Enable SFTP	Disabled
Enable network file system v3	Disabled
Allow cross-tenant replication	Disabled
Access tier	Cold
Managed Identity for SMB	Disabled

Azure Files

Require Encryption in Transit for SMB	Enabled
---------------------------------------	---------

Security

Secure transfer	Enabled
Blob anonymous access	Disabled
Allow storage account key access	Enabled
Default to Microsoft Entra authorization in the Azure portal	Disabled
Minimum TLS version	Version 1.2
Permitted scope for copy operations (Preview)	From any storage account
Enable Defender for Storage	Disabled

Networking

Public network access	Enabled
Public network access scope	Enabled from all networks
Default routing tier	Microsoft network routing

Data protection

Point-in-time restore	Disabled
Blob soft delete	Enabled
Blob retainment period in days	7
Container soft delete	Enabled
Container retainment period in days	7
Classic file share soft delete	Enabled
Classic file share retainment period in days	7
Versioning	Disabled
Blob change feed	Disabled
Version-level immutability support	Disabled

Encryption

Encryption type	Microsoft-managed keys (MMK)
Enable support for customer-managed keys	Blobs and files only
Enable infrastructure encryption	Disabled

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Create

Home



pipeline3sa_1782288421299 | Overview

Deployment

Search



Delete



Cancel



Redeploy



Download



Refresh

Overview

Inputs

Outputs

Template

Deployment is in progress

Deployment details

Resource	Type	Status	Operation de
No results.			

Give feedback

Tell us about your experience with deployment

Successfully created blob storage account

Home



pipeline3sa_1782288421299 | Overview

Deployment

Search



Delete



Cancel



Redeploy



Download



Refresh

Overview

Inputs

Outputs

Template

Your deployment is complete



Deployment name: pipeline3sa_1782288421299
Subscription: [NK](#)
Resource group: [pipeline3rg](#)

Start time: 24/6/2026, 1:39:10 pm

Correlation ID: 78912773-8a7f-41a9-9028-7c0e18f295c1



Deployment details

Next steps

Go to resource

Give feedback

Go to resource---storage browser---blob container(right side menu)---create a container---add container---data-upload the file

Home > pipeline3sa_1782288421299 | Overview > pipeline3sa

pipeline3sa | Storage browser Help me save costs by tiering unused blobs

Storage account

Search

- Overview
- Activity log
- Tags
- Diagnose and solve problems
- Access Control (IAM)
- Data migration
- Events
- Storage browser**
- Storage Mover
- Partner solutions
- Resource visualizer

pipeline3sa

- Favorites
- Recently viewed
- Blob containers
 - Slogs
 - data
- View all
- Classic file shares
- Queues
- Tables

+ Add Directory ↑ Upload 🔒 Change access le

Blob containers > data

Authentication method: Access key ([Switch to Microsoft Entr](#))

Add filter

Search blobs by prefix (case-sensitive)

Showing all 1 items

Name	Last modified
Emp1.csv	24/06/2026, 13:42:37

2)TO CREATE ADLS ACCOUNT:

*****Note:

Only difference is from the blob storage (we don't enable the hierarchical name space) but in ADLS(We enable hierarchical name space)

2.1)Create Resource:

Azure services

2.2) Click on the + icon and select the storage account:

Home

Create a resource I need a new low-cost VM

Get Started

Recently created

Categories

- AI Apps and Agents
- Machine Learning
- Analytics
- Blockchain
- Infrastructure Services
- Databases
- Developer Tools
- DevOps
- Identity
- Integration
- Internet of Things
- IT & Management Tools

Search services and marketplace

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Key Vault Create

Virtual machine Create

Storage account **Create**

Logic App Create

Ubuntu Pro 24.04 LTS Create | Learn more

Free SQL Server License: SQL Server 2022 Developer on ... Create | Learn more

Red Hat Enterprise Linux10 (latest minor version) Create | Learn more

Debian 13 "Trixie" Create | Learn more

Give feedback

2.3)Fill all the basic details:

Subscription: Give your subscription :ex)Azure subscription1

Resource group: click create---new-Create a new resource group

Instance details:

Storage account name: give any name for storage account like pipelin3adlsa

Region: select your nearest region

Preferred storage type: Azure blob storage and azure data lake storage gen2

Performance: Standard

Redundancy: LRS depends on the data

Home > [Create a resource](#)

Create a storage account ...

Select the subscription in which to create the new storage account. Choose a new or existing resource group to organize and manage your storage account together with other resources.

Subscription *	NK
Resource group *	pipeline3rg Create new

Instance details

Storage account name * ⓘ	pipeline3adlsa
Region * ⓘ	(US) West US 3 Deploy to an Azure Extended Zone
Preferred storage type	Choose preferred storage type This helps us provide relevant guidance. It doesn't restrict your storage to this resource type. Learn more
Performance * ⓘ	☒ Standard: Recommended for most scenarios (general-purpose v2 account) ☐ Premium: Recommended for scenarios that require low latency.
Redundancy * ⓘ	Locally redundant storage (LRS)

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[Review + create](#)

Fill in the advanced details:

Blob storage:select how often we use the data depends on(HOT,COOL,COLD)

Public network access:Enable

Public network access scope:Enable all networks

Routing preference:Microsoft Network Routing

*****Hierarchical Namespace:Enable

Enable soft delete:Give the no of days you can retrive the data once it deleted

Create a storage account ...

- Basics
- Advanced
- Networking
- Data protection
- Security
- Encryption
- Tags
- Review + create

Hierarchical Namespace

Hierarchical namespace, complemented by Data Lake Storage Gen2 endpoint, enables file and directory semantics, accelerates big data analytics workloads, and enables access control lists (ACLs) [Learn more](#)

Enable hierarchical namespace 

[View automation template](#)

Basics

Subscription	NK
Resource group	pipeline3rg
Location	West US 3
Storage account name	pipeline3adlsa
Primary service	
Performance	Standard
Replication	Locally redundant storage (LRS)

Advanced

Enable hierarchical namespace	Enabled
Enable SFTP	Disabled
Enable network file system v3	Disabled
Allow cross-tenant replication	Disabled
Access tier	Cold
Managed Identity for SMB	Disabled

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Home > Create a resource

Create a storage account

Basics Advanced Networking Data protection Security Encryption Tags Review + create

[View automation template](#)

Basics

Subscription	NK
Resource group	pipeline3rg
Location	West US 3
Storage account name	pipeline3adlsa
Primary service	
Performance	Standard
Replication	Locally redundant storage (LRS)

Advanced

Enable hierarchical namespace	Enabled
Enable SFTP	Disabled
Enable network file system v3	Disabled
Allow cross-tenant replication	Disabled
Access tier	Cold
Managed Identity for SMB	Disabled

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[Give feedback](#)

*** **Submitting deployment...**
Submitting the deployment template for resource 'pipeline3rg'.

Home



pipeline3adlsa_1782289224022 | Overview

Deployment

[Delete](#) [Cancel](#) [Redeploy](#) [Download](#) [Refresh](#)

Overview

[Inputs](#)

[Outputs](#)

[Template](#)

Deployment is in progress



Deployment name: pipeline3adlsa_1782289224022
Subscription: [NK](#)
Resource group: [pipeline3rg](#)

Start time: 24/1
Correlation ID:

Deployment details

Resource	Type
pipeline3adlsa	Microsoft.Storage/storageAccounts

Give feedback

[Tell us about your experience with deployment](#)

Successfully adls storage account is also created

Home



pipeline3adlsa_1782289224022 | Overview ...

Search

Delete Cancel Redeploy Download Refresh

Overview

Inputs
 Outputs
 Template

Your deployment is complete



Deployment name: pipeline3adlsa_1782289224022
Subscription: [NK](#)
Resource group: [pipeline3rg](#)

Start time: :
Correlation

Deployment details

Next steps

[Go to resource](#)

Give feedback

[Tell us about your experience with deployment](#)

Click on Go to resource---storage browser---Blob containers---add container---we can add directory---can upload the data

Home > pipeline3adlsa_1782289224022 | Overview > pipeline3adlsa

pipeline3adlsa | Storage browser ... [Help me save costs by tiering unused blobs](#)

Search

pipeline3adlsa

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Partner solutions
Resource visualizer
Data storage
Security + networking

pipeline3adlsa

Favorites
Recently viewed
Blob containers
Blogs
practice
View all
Classic file shares
Queues
Tables

+ Add Directory Upload Refresh Delete Copy Paste Rename Acquire lease Break lease Edit columns

[Blob containers](#) > [practice](#) > [dataset](#)

Authentication method: Access key ([Switch to Microsoft Entra user account](#))

Search blobs by prefix (case-sensitive)

Only show active objects

Showing all 1 items

☐	Name	Last modified	Access tier	Blob type	Size	Lease state
☐	[...]					
☐	training11111	24/06/2026, 13:54:48	Cold (inferred)	Block blob	25.73 KiB	Available

3)TO CREATE AZURE SQL DATABASE

Azure services



Create a resource



Resource groups



Cost Management



Subscriptions



Quickstart Center



Foundry



Kubernetes services



Virtual machines



App Services



More services

Click+---to create resource---select the database and create

Home

Create a resource



I need a new low-cost VM

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Azure SQL
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Azure Cosmos DB
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Azure Database for PostgreSQL Flexible Server
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PostgreSQL on Ubuntu 24.04
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Percona Server for MySQL 5.7 on Ubuntu
[Create](#) | [Learn more](#)

Fill in all the basic details:

- Click "Create a resource" (top left)
- Search for "SQL Database", then click Create
- Fill in the basics:
 - o Subscription: Choose your subscription
- Resource Group: Select the one you created
- Database name: Give database name
- Server: Click "Create new"
- **Create a SQL Server :**
 - This is where your database will present.
 - Click Create new (under SQL Server)
- Server name: (must be unique)
- Location: Same as resource group
- Admin login: Create a username
- Password: Set a strong password o Click OK to return to the SQL DB creation form.
- **Configure Database(DTU)Basic**
- **Enable Public Access (Firewall Rule):**
 - Connectivity Method: Public Endpoint
 - In the Networking tab, under Firewall rules, select: Allow Azure services and resources to access this server: yes
- Click Review + create
- Wait for validation
- Click Create

Create SQL Database ...

Microsoft

- Basics
- Networking
- Security
- Additional settings
- Tags
- Review + create

Product details

SQL database
by Microsoft
[Terms of use](#) | [Privacy policy](#)

Estimated cost per month
4.90 USD

Terms

By clicking "Create", I (a) agree to the legal terms and privacy statement(s) associated with the Marketplace listed above; (b) authorize Microsoft to bill my current payment method for the fees associated with the the same billing frequency as my Azure subscription; and (c) agree that Microsoft may share my contact transactional information with the provider(s) of the offering(s) for support, billing and other transactional Microsoft does not provide rights for third-party offerings. For additional details see [Azure Marketplace](#)

Basics

Subscription	NK
Resource group	pipeline3rg
Region	West US 2
Database name	pipeline3db
Server	(new) pipeline3server
Authentication method	SQL authentication

- Create
- < Previous
- [Download a template for automation](#)

Home

Microsoft.SQLDatabase.newDatabaseNewServer_3b9f0fc844644a4aa6b3a | Overview ...

Deployment

Search

Delete

Cancel

Redeploy

Download

Refresh

Overview

Inputs

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Template

Deployment is in progress

Deployment name : Microsoft.SQLDatabase.newDatabaseNewServer_3b9f0fc844644a4aa6b3a Start time : 24/6/2026, 2:04:26 pm
Subscription : NK Correlation ID : 9ceab9d7-b6e1-4809-8e6e-3dc9db4f9e3c
Resource group : pipeline3rg

Deployment details

Resource	Type	Status
pipeline3server	Microsoft.Sql/servers	Accepted

Succesfully created Azure sql db

Home

Microsoft.SQLDatabase.newDatabaseNewServer_3b9f0fc844644a4aa6b3a | Overview ...

Deployment

Search x << Delete Cancel Redeploy Download Refresh

Overview

Inputs

Outputs

Template

✓ Your deployment is complete

Deployment name : Microsoft.SQLDatabase.newDatabaseNewServer_3b9f0fc844644a4aa6b3a Start time : 24/6/2026, 2:04:27 pm
 Subscription : NK Correlation ID : 9ceab9d7-b6e1-4809-8e6e-3dc9db4f9e3c
 Resource group : pipeline3rg

> Deployment details

Next steps

[Go to resource](#)

Click on the go to resource---query editor(preview)---give sql authentication(username,password)

Home > Microsoft.SQLDatabase.newDatabaseNewServer_3b9f0fc844644a4aa6b3a | Overview > pipeline3db (pipeline3server/pipeline3db)

pipeline3db (pipeline3server/pipeline3db) | Query editor (preview) ☆ ...

SQL database

Search ○ << New query Templates Open in Feedback Sign out Classic experience

Overview

Activity log

Tags

Diagnose and solve problems

Query editor (preview)

Mirror database in Fabric (preview)

Resource visualizer

> Settings

> Data management

> Integrations

> Power Platform

> Security

> Intelligent performance

> Monitoring

> Automation

> Help

Explorer <<

Search

pipeline3db

> dbo

> Queries



Start exploring

Write and run SQL queries, create database objects, and more. [View documentation](#)

[New query](#)

4)TO CREATE ADF:

Azure services



Home

Create a resource

...



I need a new low-cost VM

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Categories

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...

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Microsoft Fabric
[Create](#)



Azure Databricks
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Data Factory
[Create](#)



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[Create](#)



Azure Machine Learning
[Create](#)

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SUSE Linux Enterprise Server for
[Create](#) | [Learn more](#)



SQL Server 2022 Reporting Services
[Create](#) | [Learn more](#)



Cisco Catalyst 8000V Edge Software
[Create](#) | [Learn more](#)



MongoDB Products & Services
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Fill in the basic details required:

SubscriptionName:Give your subscription

Resource group: Pipeline3rg

Name: pipeline3adfbike

Region: select region

Version: V2

Networking:

Connect via:Public endpoint

Create Data Factory ...

- Basics
- Git configuration
- Networking
- Advanced
- Tags
- Review + create

 [View automation template](#)

Basics

Subscription	NK
Resource group	pipeline3rg
Name	pipeline3adfbike
Region	East US
Version	V2

Networking

Connect via	Public endpoint
-------------	-----------------

Previous

Next

Create

Click on create---wait for deployment

Sucessfully created ADF

C

Home



Microsoft.DataFactory-20260624141224 | Overview

Deployment

Search



Delete



Cancel



Redeploy



Download



Refresh

Overview

Inputs

Outputs

Template



Your deployment is complete



Deployment name : Microsoft.DataFactory-20260624141224

Subscription : [NK](#)

Resource group : [pipeline3rg](#)



Deployment details



Next steps

[Go to resource](#)

Give feedback



[Tell us about your experience with deployment](#)

Go to resource ---click on the data factory---launch azure data studio



pipeline3adfbike

Data factory (V2)



Search

Delete

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Resource visualizer

Settings

Getting started

Monitoring

Automation

Help

Essentials

Resource group ([move](#)) : [pipeline3rg](#)

Status : Succeeded

Location : East US

Subscription ([move](#)) : [NK](#)

Subscription ID : 5bd8f1cd-03ad-48fa-8410-6e58274fc11f

Type : Data factory (V2)

Getting started : [Quick start](#)



Azure Data Factory Studio

[Launch studio](#)

Quick Starts



Tutorials

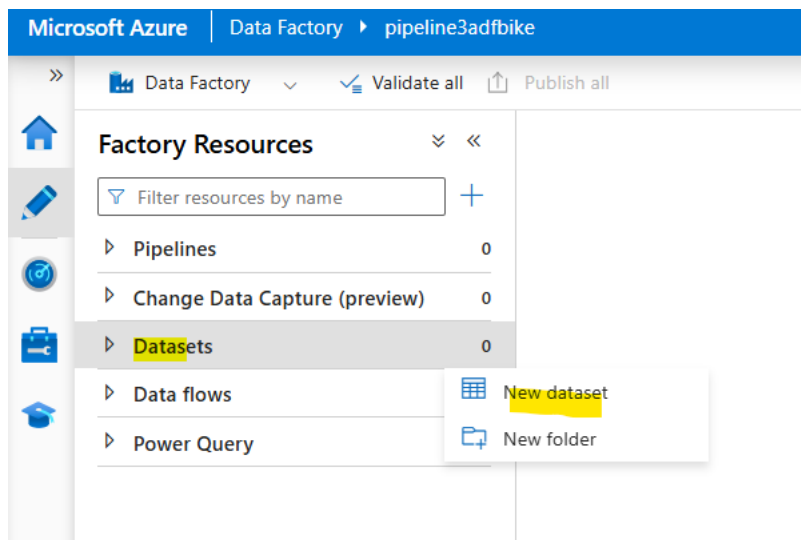


Template Gallery



Training Modules

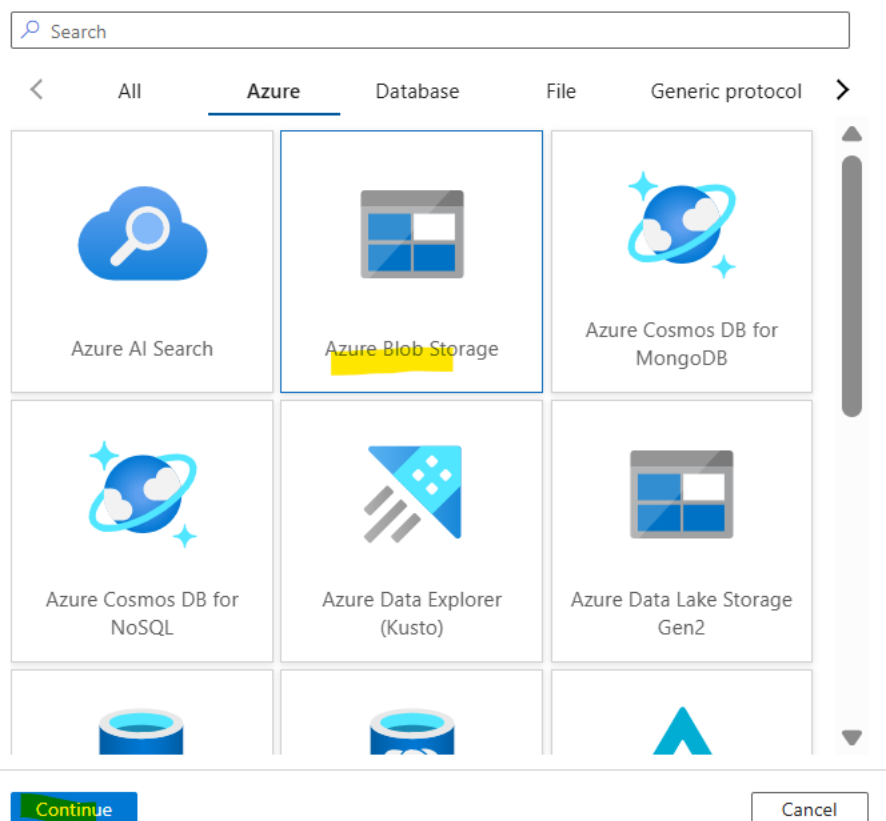
We need to copy the raw data from the storage accounts(blob and adls)



New dataset

In pipeline activities and data flows, reference a dataset to specify the location and structure of your data within a data store. [Learn more](#)

Select a data store



Create a new linked service to copy data from blob 2 adf---give subscription---storage account---test connection

New linked service

 Azure Blob Storage [Learn more](#)

Authentication type

Account key

Connection string

[Azure Key Vault](#)

Account selection method ⓘ

☒ From Azure subscription ☐ Enter manually

Azure subscription ⓘ

NK (5bd8f1cd-03ad-48fa-8410-6e58274fc11f)

Storage account name *

pipeline3sa

Additional connection properties

+ New

Test connection ⓘ

☒ To linked service ☐ To file path

Annotations

+ New

> Parameters

Create

Cancel

✓ Connection successful

 Test connection

Now provide the file path

Set properties

Name

emp_data

Linked service *

blob2adf

File path

data

/ Directory

/ Emp1.csv

First row as header



Import schema

☒ From connection/store ☐ From sample file ☐ None

OK

Back

Cancel

Preview the data to confirm whether the data is loaded or not

Preview data

Linked service: blob2adf

Object: Emp1.csv

	EmployeeID	ManagerID	Title	MaritalStatus	Gender	HireDate	Dept	Job Grade
1	1	16	Gustavo Achong	M	M	02-02-2013 00:00	Sales	Admin
2	2	6	Catherine Abel	S	M	31-08-2013 00:00	Sales	Management
3	3	12	Kim Abercrombie	M	M	16-06-2014 00:00	Finance	Admin
4	4	3	Humberto Acevedo	S	M	10-07-2014 00:00	Logistics	Admin
5	5	263	Pilar Ackerman	M	M	16-07-2014 00:00	Human Resource	Admin
6	6	109	Frances Arams	S	M	25-07-2014	Sales	Management

Connection

Linked service *

File path *

Compression type

Column delimiter

Row delimiter

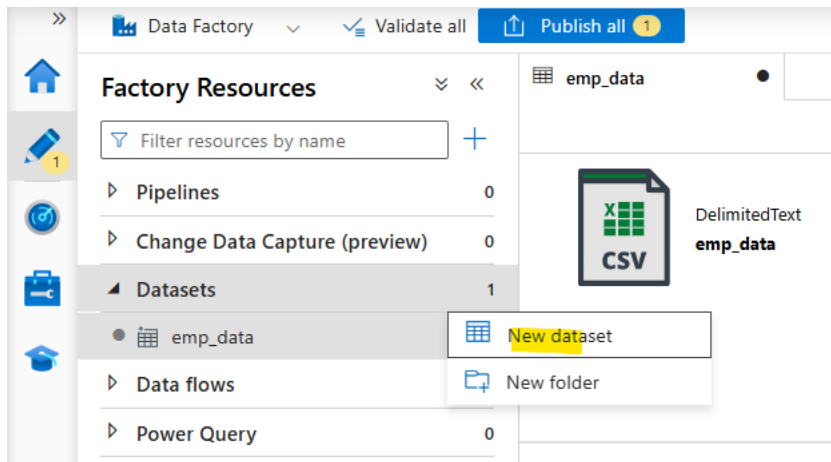
Encoding

Quote character

Escape character

Backslash (\)

***we need to create one more dataset because the data we are getting from 2 different sources

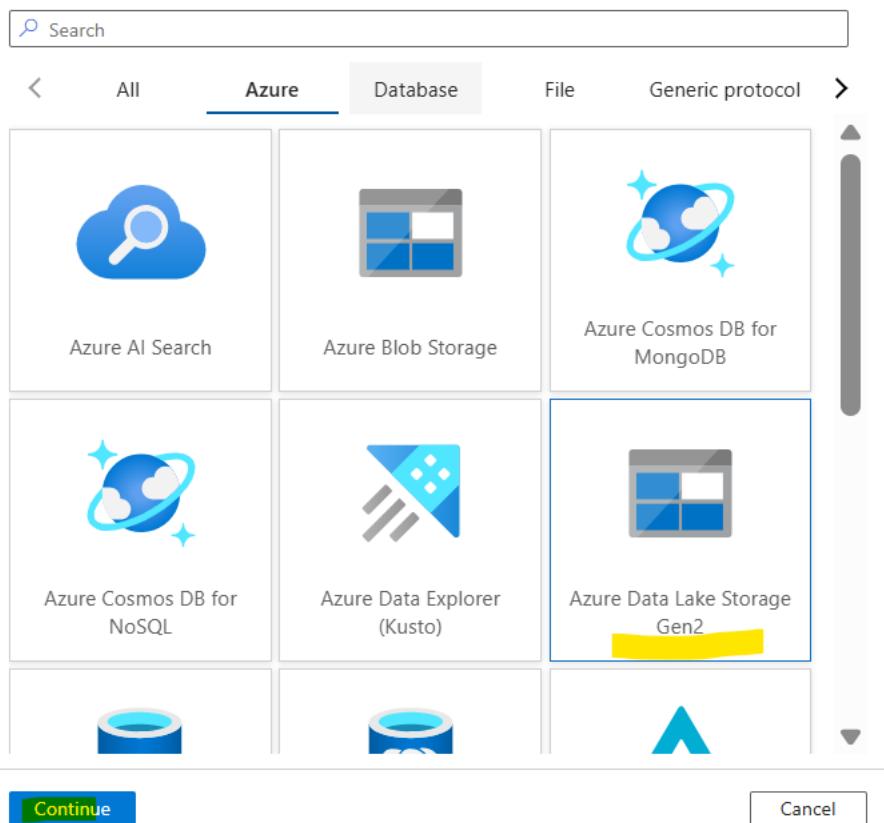


Select the storage account(adls) and the file type

New dataset

In pipeline activities and data flows, reference a dataset to specify the location and structure of your data within a data store. [Learn more](#)

Select a data store



Select format

Choose the format type of your data

 Avro	 Binary	 DelimitedText
 Excel	 Iceberg	 JSON
 ORC	 Parquet	 XML

Continue

Back

Cancel

Create a new linked service to copy data from blob---give subscription---storage account---test connection

New linked service

 Azure Data Lake Storage Gen2 [Learn more](#)

Connect via integration runtime * ⓘ

✓ AutoResolveIntegrationRuntime

Authentication type

Account key

Account selection method ⓘ

☒ From Azure subscription ☐ Enter manually

Azure subscription ⓘ

NK (5bd8f1cd-03ad-48fa-8410-6e58274fc11f)

Storage account name *

pipeline3adlsa



Test connection ⓘ

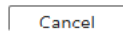
☒ To linked service ☐ To file path

Annotations

+ New

> Parameters

> Advanced ⓘ

✓ Connection successful

 Test connection

Provide the file path

Set properties

Name

Training_data

Linked service *

adls2adf

File path

practice

/ dataset

/ Training1 (1).csv

First row as header



Import schema

☒ From connection/store ☐ From sample file ☐ None

OK

Back

Cancel

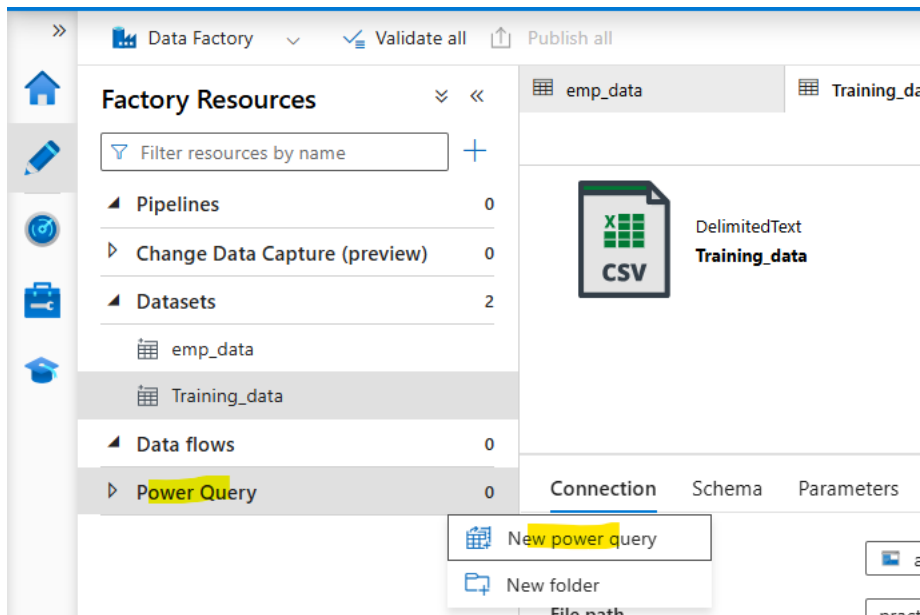
Preview the data, to see the data is loaded correctly or not

The screenshot shows the Microsoft Azure Data Factory interface. On the left, the 'Factory Resources' pane lists 'Pipelines' (0), 'Change Data Capture (preview)' (0), 'Datasets' (2), 'Data flows' (0), and 'Power Query' (0). The 'Datasets' section is expanded, showing 'emp_data' and 'Training_data'. The 'Training_data' dataset is selected. The main pane displays the 'Preview data' window for the 'Training_data' dataset, linked to the 'adls2adf' service. The object is 'Training1 (1).csv'. The preview shows a table with 7 rows and 7 columns: Start Date, Course Code, Course Name, Employee_ID, Cost, and Supplier. The data is as follows:

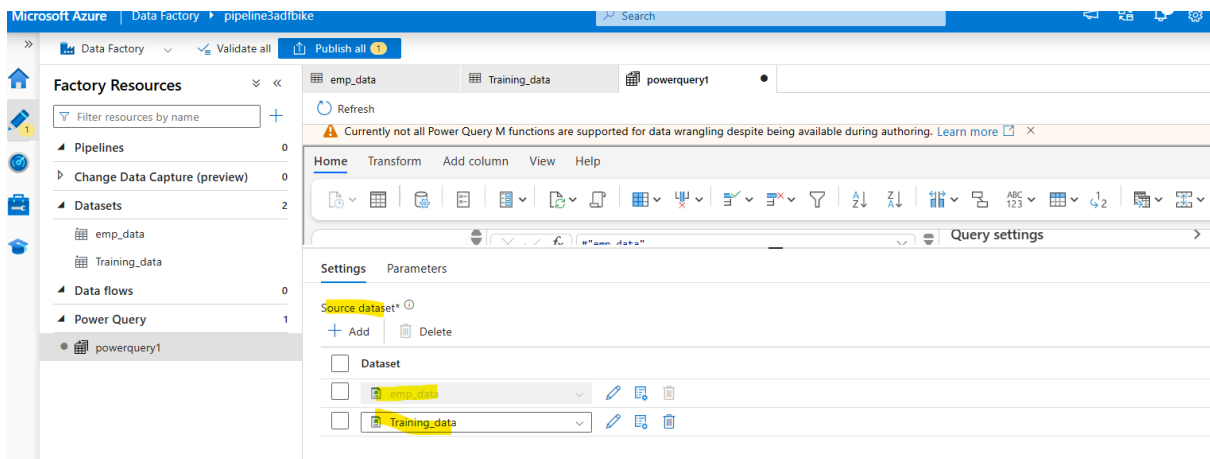
	Start Date	Course Code	Course Name	Employee_ID	Cost	Supplier
1	18-09-2015	1	Communication Workshop	1	1300	Communication Experts
2	18-09-2015	1	Communication Workshop	4	1300	Communication Experts
3	18-09-2015	1	Communication Workshop	25	1300	Communication Experts
4	18-09-2015	1	Communication Workshop	33	1300	Communication Experts
5	18-09-2015	1	Communication Workshop	44	1300	Communication Experts
6	18-09-2015	1	Communication Workshop	23	1300	Communication Experts
7	18-09-2015	1	Communication Workshop	6	1300	Communication Experts

*****Publish all the changes to adf(to avoid data loss)

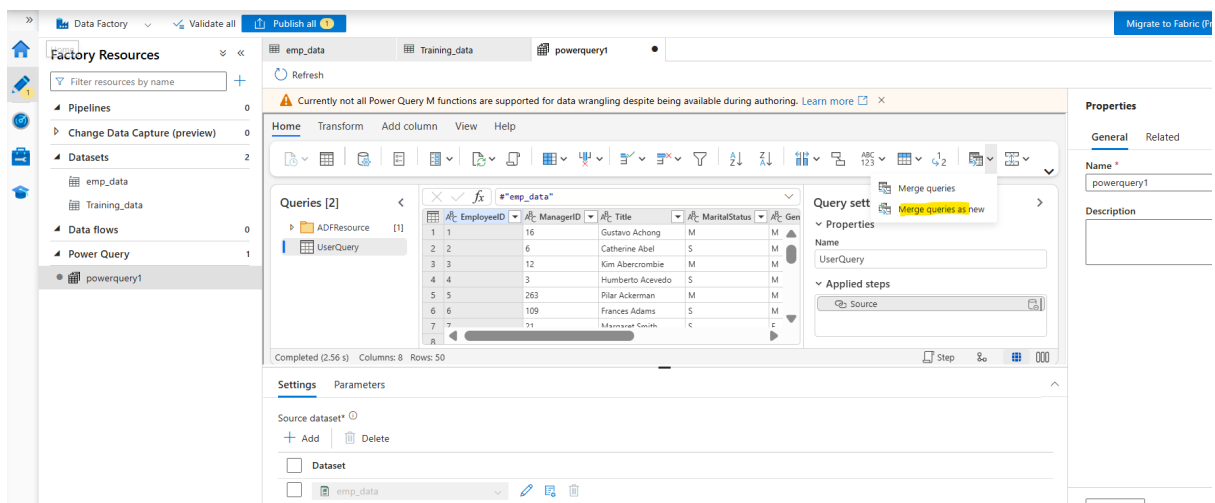
Create Powerquery for transformations:



In source we need to add the datasets



To join both the tables emp and training data in powerquery we can use merge



We need to give the left table, right table --- then select the join --- select the common columns to apply merge

Merge

Select tables and matching columns to create a merged table.

Left table for merge *

emp_data

EmployeeID	ManagerID	Title	MaritalStatus	Gender	HireDate
1	16	Gustavo Achong	M	M	02-02-2013 00:00:00
2	6	Catherine Abel	S	M	31-08-2013 00:00:00

 The selection matches 47 rows from both the tables

OK

Cancel

Merge

Select tables and matching columns to create a merged table.

Training_data

Start Date	Course Code	Course Name	Employee_ID	Cost	Supplier
18-09-2015	1	Communication Works...	1	1300	Communica
18-09-2015	1	Communication Works...	4	1300	Communica
18-09-2015	1	Communication Works...	25	1300	Communica

 The selection matches 47 rows from both the tables

OK

Cancel

Merge

Select tables and matching columns to create a merged table.


Left outer


Right outer


Full outer


Inner


Left anti


Right anti

☐ Use fuzzy matching to perform the merge

 The selection matches 47 rows from both the tables

OK

Cancel

Both tables got merged, click on the icon to extract the required columns

Home Transform Add column View Help

Queries [4]

- ADFResource [2]
 - UserQuery
 - Merge

Table.NestedJoin(emp_data, {"EmployeeID"}, Training_data, ...)

	Gender	HireDate	Dept	Job Grade	Training_data
1	M	02-02-2013 00:00	Sales	Admin	[Table]
2	M	31-08-2013 00:00	Sales	Management	[Table]
3	M	10-07-2014 00:00	Logistics	Admin	[Table]
4	M	16-06-2014 00:00	Finance	Admin	[Table]
5	M	09-07-2015 00:00	Production	Operations	[Table]
6	M	13-07-2015 00:00	Sales	Operations	[Table]
7	M	16-07-2014 00:00	Human Resour...	Admin	[Table]
8	M	18-07-2015 00:00	Sales	Operations	[Table]
9	M	25-07-2014 00:00	Sales	Management	[Table]
10	M	08-07-2015 00:00	Human Resour...	Management	[Table]
11	F	31-07-2014 00:00	Sales	Operations	[Table]
12	F	11-08-2014 00:00	Finance	Management	[Table]
13	M	12-08-2014 00:00	Logistics	Admin	[Table]

Query settings

Properties

Name

Merge

Applied steps

Source

Home Transform Add column View Help

Queries [4]

- ADFResource [2]
 - UserQuery
 - Merge

Table.NestedJoin(emp_data, {"EmployeeID"}, Training_data, ...)

	Gender	HireDate	Dept	Job Grade	Training_data
1	M	02-02-2013 00:00	Sales	Admin	[Table]
2	M	31-08-2013 00:00	Sales	Management	[Table]
3	M	10-07-2014 00:00	Logistics	Admin	[Table]
4	M	16-06-2014 00:00	Finance	Admin	[Table]
5	M	09-07-2015 00:00	Production	Operations	[Table]
6	M	13-07-2015 00:00	Sales	Operations	[Table]
7	M	16-07-2014 00:00	Human Resour...	Admin	[Table]
8	M	18-07-2015 00:00	Sales	Operations	[Table]
9	M	25-07-2014 00:00	Sales	Management	[Table]
10	M	08-07-2015 00:00	Human Resour...	Management	[Table]
11	F	31-07-2014 00:00	Sales	Operations	[Table]
12	F	11-08-2014 00:00	Finance	Management	[Table]
13	M	12-08-2014 00:00	Logistics	Admin	[Table]

Query settings

Properties

Search

(Select all)

Start Date

Course Code

Course Name

Employee_ID

Cost

Supplier

Use original column name as prefix

OK Cancel

Lets see the data

Factory Resources

Filter resources by name

Pipelines 0

Change Data Capture (pr... 0

Datasets 2

- emp_data
- Training_data

Data flows 0

Power Query 1

- powerquery1

emp_data Training_data powerquery1

Refresh

Currently not all Power Query M functions are supported for data wrangling despite being available during authoring. [Learn more](#)

Home Transform Add column View Help

Queries [4]

- ADFResource [2]
 - UserQuery
 - Merge

Table.ExpandTableColumn(Source, "Training_data", {"Start Date", "Course Code", "Course Name", ...})

	Gender	HireDate	Dept	Job Grade	Start Date	Course Code	Course Name
1	M	02-02-2013 00:00	Sales	Admin	18-09-2015	1	Communication Workshop
2	M	31-08-2013 00:00	Sales	Management	22-01-2017	2	Communication Workshop
3	M	10-07-2014 00:00	Logistics	Admin	18-09-2015	1	Communication Workshop
4	M	16-06-2014 00:00	Finance	Admin	16-01-2018	3	Communication Workshop
5	M	09-07-2015 00:00	Production	Operations	12-08-2015	6	Instituting Disciplinary Action
6	M	13-07-2015 00:00	Sales	Operations	18-09-2015	1	Communication Workshop
7	M	13-07-2015 00:00	Sales	Operations	04-09-2015	4	Communication Workshop
8	M	16-07-2014 00:00	Human Resour...	Admin	22-01-2017	2	Communication Workshop
9	M	18-07-2015 00:00	Sales	Operations	08-09-2016	5	Advanced Excel
10	M	18-07-2015 00:00	Sales	Operations	18-09-2015	1	Communication Workshop
11	M	18-07-2015 00:00	Sales	Operations	04-09-2015	4	Communication Workshop
12	M	18-07-2015 00:00	Sales	Operations	08-09-2016	5	Advanced Excel
13	M	25-07-2014 00:00	Sales	Management	18-09-2015	1	Communication Workshop
14	M	08-07-2015 00:00	Human Resour...	Management	18-09-2015	1	Communication Workshop

Query setting

Properties

Name

Merge

Applied steps

Source

Completed (4.98 s) Columns: 13 Rows: 99+

Settings Parameters

Cancel

Publish all the changes in the adf before creating a pipeline

Microsoft Azure | Data Factory | pipeline3adfbike

Factory Resources

- Pipelines: 0
- Change Data Capture (pr...): 0
- Datasets: 2
 - emp_data
 - Training_data
- Data flows: 0
- Power Query: 1
 - powerquery1

emp_data Training_data powerquery1

Refresh

Currently not all Power Query M functions are supported for data wrangling despite being available during authoring. [Learn more](#)

Home Transform Add column View Help

Queries [4]

- ADFRE... [2]
- UserQuery
- Merge

Table.ExpandTableColumn(Source, "Training_data", {"Start Date", "Course Code", "Course Name",

	Gender	HireDate	Dept	Job Grade	Start Date	Course Code	Course Name
1	M	02-02-2013 00:00	Sales	Admin	18-09-2015	1	Communication Workshop
2	M	31-08-2013 00:00	Sales	Management	22-01-2017	2	Communication Workshop
3	M	10-07-2014 00:00	Logistics	Admin	18-09-2015	1	Communication Workshop
4	M	10-07-2014 00:00	Logistics	Admin	16-01-2018	3	Communication Workshop
5	M	16-06-2014 00:00	Finance	Admin	12-08-2015	6	Instituting Disciplinary Action
6	M	09-07-2015 00:00	Production	Operations	18-09-2015	1	Communication Workshop
7	M	13-07-2015 00:00	Sales	Operations	18-09-2015	1	Communication Workshop
8	M	13-07-2015 00:00	Sales	Operations	04-09-2015	4	Communication Workshop
9	M	16-07-2014 00:00	Human Resour...	Admin	22-01-2017	2	Communication Workshop
10	M	16-07-2014 00:00	Human Resour...	Admin	08-09-2016	5	Advanced Excel
11	M	18-07-2015 00:00	Sales	Operations	18-09-2015	1	Communication Workshop
12	M	18-07-2015 00:00	Sales	Operations	04-09-2015	4	Communication Workshop
13	M	18-07-2015 00:00	Sales	Operations	08-09-2016	5	Advanced Excel
14	M	25-07-2014 00:00	Sales	Management	18-09-2015	1	Communication Workshop
15	M	08-07-2015 00:00	Human Resour...	Management	18-09-2015	1	Communication Workshop
16							

Completed (4.98 s) Columns: 13 Rows: 99+

Settings Parameters

TO CREATE PIPELINE:

Data Factory | Validate all | Publish all 1

Factory Resources

- Pipelines: 0
 - New pipeline
 - Pipeline from template
 - New folder
- Change Data Capture (pr...): 0
- Datasets: 2
 - emp_data
 - Training_data
- Data flows: 0
- Power Query: 1
 - EMP_TRAIN_PQ

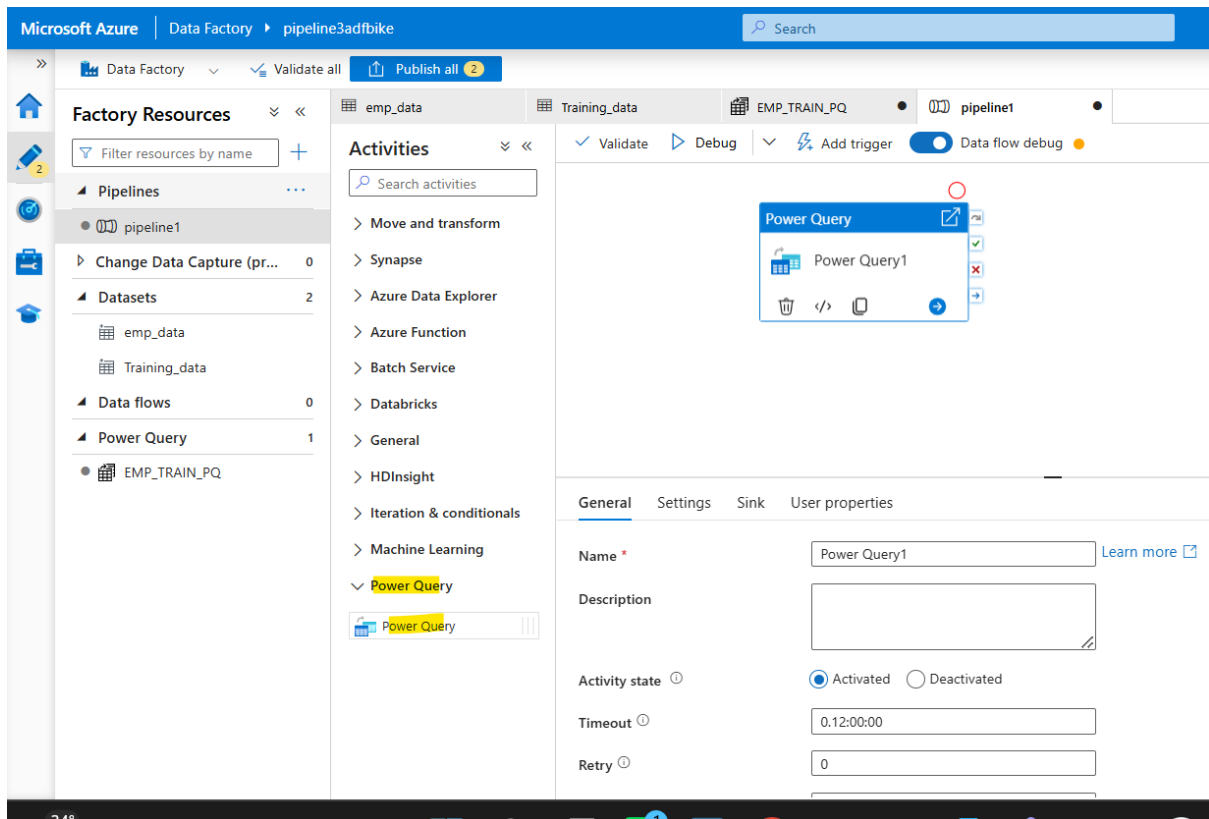
emp_data Training_data

Home Transform

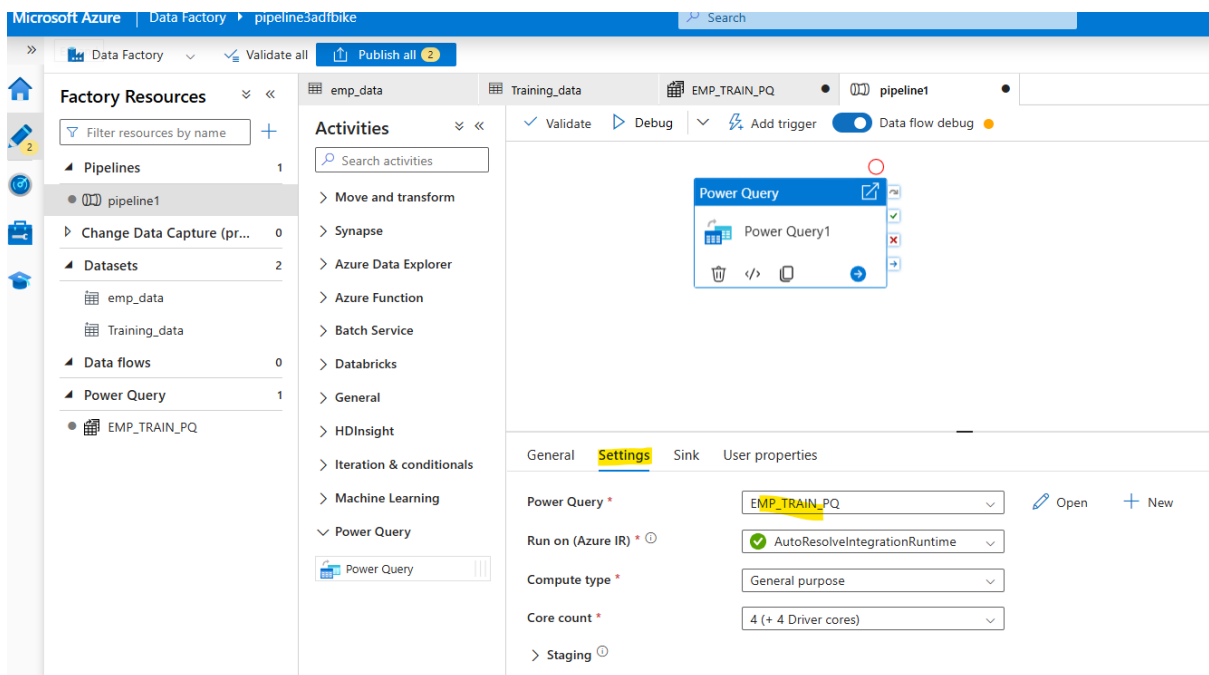
ADFRE... [2]

UserQuery

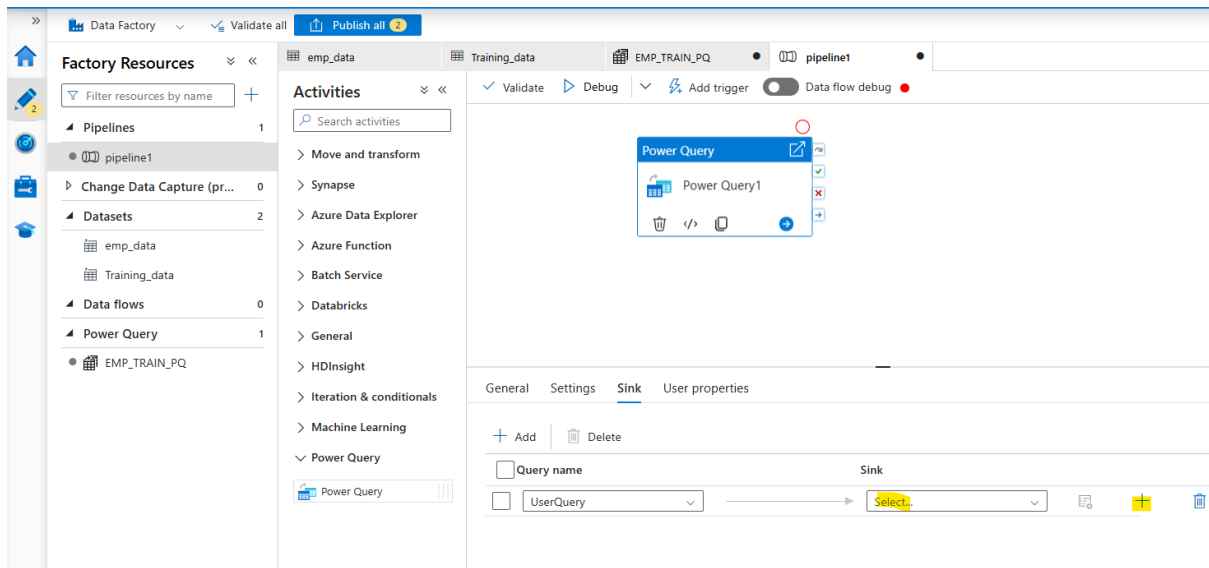
Merge



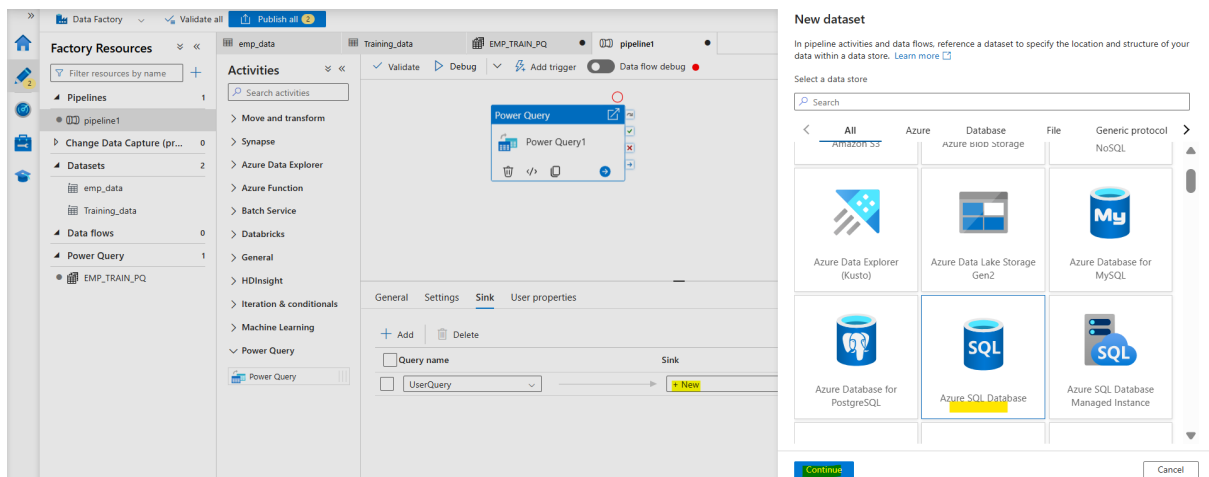
In settings add the power query where we have done the transformations



In power query we don't have any sink for the target so we can directly create sink in on top of the pipeline .



Click + to create target—select the target---create a linked service(fill all the details)-test connection---create a new table



Set properties

Name

EMPTRAINING_TB

Linked service *

adf2azsqldb

☐ Select from existing table ☒ New table

Schema and table name

Schema name

emptrain_tb

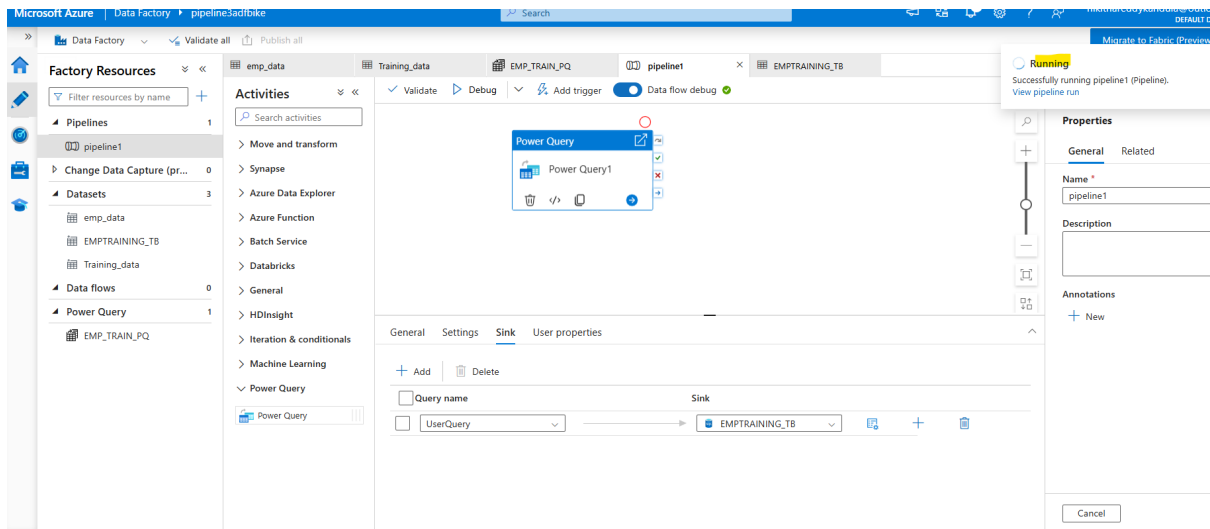
OK

Back

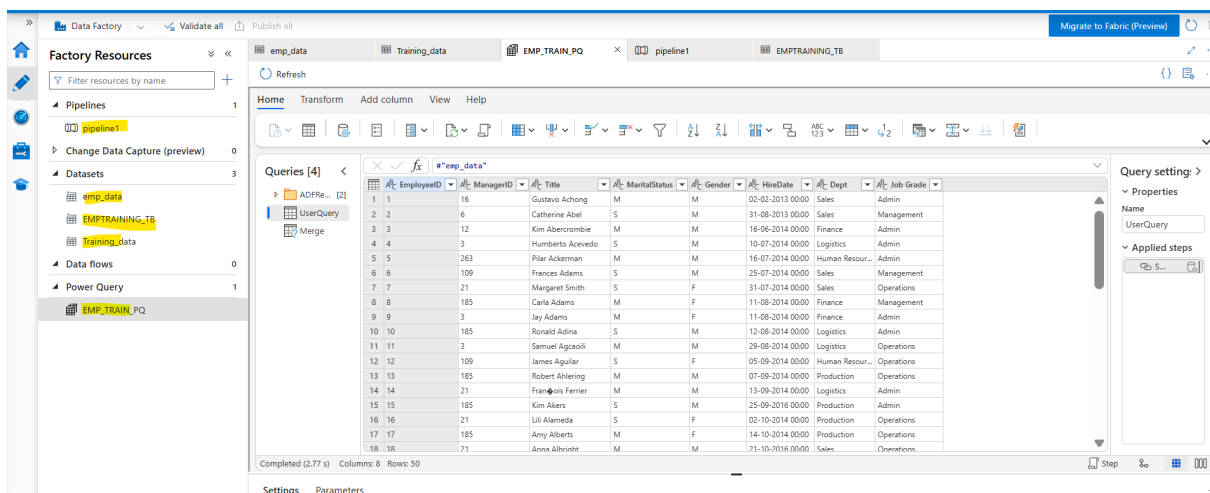
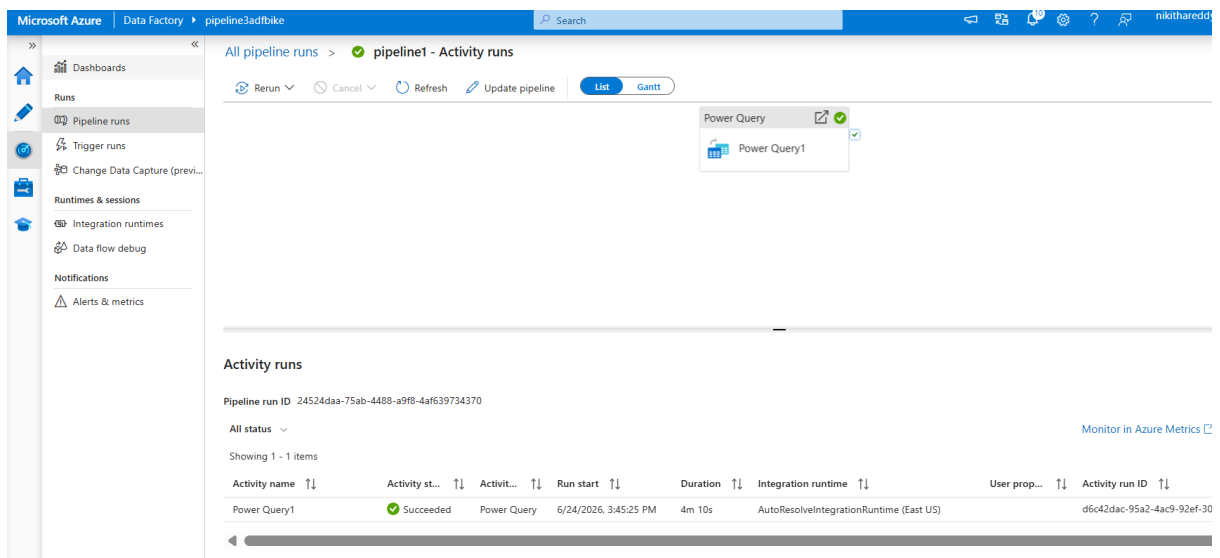
Cancel

Publish all the changes---add trigger—triggernow-run pipeline

The screenshot displays the Microsoft Azure Data Factory console. On the left, the 'Factory Resources' pane shows a tree view with 'Pipelines' (1 item), 'Datasets' (3 items: emp_data, EMPTRAINING_TB, Training_data), 'Data flows' (0 items), and 'Power Query' (1 item: EMP_TRAIN_PQ). The 'Activities' pane on the right lists various activity categories like 'Move and transform', 'Synapse', 'Azure Data Explorer', 'Azure Function', 'Batch Service', 'Databricks', 'General', 'HDInsight', 'Iteration & conditionals', 'Machine Learning', and 'Power Query'. The 'Power Query' category is expanded, showing 'Power Query' as an available activity. The main canvas shows a 'Power Query' activity named 'Power Query1' being configured. The 'Sink' tab is active, showing a 'Query name' dropdown set to 'UserQuery' and a 'Sink' dropdown set to 'EMPTRAINING_TB'. The 'Add trigger' button is visible at the top of the canvas.



Successfully the pipeline executed:



Check in the target:

Microsoft Azure

Search resources, services, and docs (G+)

Copilot

nikithareddykandula@o...
DEFAULT DIRECTORY (NORTHREAME...

Home > pipeline3db (pipeline3server/pipeline3db)

pipeline3db (pipeline3server/pipeline3db) | Query editor (preview) ☆ ...

SQL database

Search

New query Templates Open in Feedback Sign out Classic experience

Overview
Activity log
Tags
Diagnose and solve problems
Query editor (preview)
Mirror database in Fabric (preview)
Resource visualizer
Settings
Data management
Integrations
Power Platform
Security
Intelligent performance
Monitoring
Automation
Help

Explorer
Search
pipeline3db
 dbo
 Tables
 emptrain_tb
 Views
 Stored Procedures
 Functions
 Queries
 SQL query 1
 SQL query 2

emptrain_tb SQL query 2

Run Save as view

1 SELECT TOP (1000) * FROM [dbo].[emptrain_tb]

Messages Results

	EmployeeID	ManagerID	Title	MaritalStatus	Gender	HireDate	Dept	Job Grade
1	1	16	Gustavo Achong	M	M	02-02-2013 00:00	Sales	Admin
2	2	6	Catherine Abel	S	M	31-08-2013 00:00	Sales	Management
3	3	12	Kim Abercrombie	M	M	16-06-2014 00:00	Finance	Admin
4	4	3	Humberto Acevedo	S	M	10-07-2014 00:00	Logistics	Admin
5	5	263	Pilar Ackerman	M	M	16-07-2014 00:00	Human Resource	Admin
6	6	109	Frances Adams	S	M	25-07-2014 00:00	Sales	Management
7	7	21	Margaret Smith	S	F	31-07-2014 00:00	Sales	Operations
8	8	185	Carla Adams	M	F	11-08-2014 00:00	Finance	Management
9	9	3	Jay Adams	M	F	11-08-2014 00:00	Finance	Admin

Succeeded (0 sec 404 ms) Columns: 8 Rows: 50